Optimal enfranchisement

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Abstract

A planner picks the franchise — the subset of agents allowed to vote — for a binary decision that affects different agents with different intensities. We frame the planner's problem as maximizing expected stake-weighted agreement with the majority outcome, and study two natural questions: (i) is some franchise always at least as good as the planner just picking the outcome herself, and (ii) is the optimal franchise composed of the highest-stake agents? Both answers are no in full generality, even when the joint opinion distribution has full support: we give two five-agent counterexamples. Once opinions are independent and identically distributed both answers become yes, and under fair-coin opinions the optimal franchise reduces to a one-dimensional prefix rule.

1 Introduction

Many collective decisions are binary — admit a country, lift a curfew, ban a substance — and bear unequally on the population. A normative theory that takes preference intensity seriously must decide who should vote. We give a stylized model of this trade-off and ask two natural questions:

1. Is voting always at least as good as letting the planner pick an outcome ex ante?
2. Is the optimal franchise the one composed of the highest-stake agents?

We show that, under full support of the joint opinion distribution alone, the answer to both questions is no. Section 3 gives a five-agent example where every odd majority franchise is beaten by the constant rule $Y \equiv 1$, and a second five-agent example where the unique optimal franchise excludes a strictly higher-stake agent in favour of a lower-stake one. Section 4 then identifies a clean restriction — common iid opinions — under which both questions get a positive answer, and derives a closed-form prefix rule when opinions are i.i.d. fair coins.

Related literature. The closest classical results are the Condorcet jury theorems [1], which give independence and competence conditions under which majority correctness rises with the size of a homogeneous jury. The expert-voting literature [2, 3] solves for optimal weighted majority rules when voter competences are observed, with weights $\log(p_i/(1-p_i))$. Schmitz and Tröger [4] study ex-ante utilitarian welfare for binary alternatives with uncertain preference intensities; their main message agrees with the one here — symmetry and neutrality make majority attractive, independence restores strong optimality, and correlated utilities can make majority suboptimal. The Penrose–Banzhaf voting-power tradition [5, 6] stresses that a voter's marginal influence is governed by the probability of a tie among the others, a central binomial coefficient; this is the same quantity that drives the insider advantage in our positive results. For schemes that aggregate intensity directly rather than choosing a franchise, see Casella and Gelman [7].

2 Model

A planner faces a population $\mathcal{N} = \{1, \dots, N\}$ and a binary decision $\{0, 1\}$. From her vantage, agent i's opinion is a Bernoulli random variable X_i. The planner picks a *franchise* $C \subseteq \mathcal{N}$ and the outcome is the strict majority within C:

$$Y_C = \mathbf{1}_{\{S_C > |C|/2\}}, \qquad S_C = \sum_{i \in C} X_i.$$

Strict majority is unbiased only when $|C|$ is odd, so we assume throughout that N is odd and consider only nonempty franchises of odd size. Let $p(i, C) = P[X_i = Y_C]$.

Each agent has utility $U(i, C)$ that depends on her own success.
Assumption 1 (Selfishness)**.** Conditional on $\mathbf{1}_{\{X_i = Y_C\}}$, agent i is indifferent to the success of others and to anyone's inclusion in the franchise.

Under Assumption 1, $\mathbb{E}[U(i, C) \mid X_i = Y_C]$ and $\mathbb{E}[U(i, C) \mid X_i \neq Y_C]$ are constants in C. Define

$$\Delta_i = \mathbb{E}[U(i, C) \mid X_i = Y_C] - \mathbb{E}[U(i, C) \mid X_i \neq Y_C] \geq 0,$$

the agent's *stake*. Label agents so $\Delta_1 \geq \Delta_2 \geq \cdots \geq \Delta_N \geq 0$ and drop the constant terms from utility. Aggregate expected welfare under franchise C is

$$U(C) = \sum_{i \in \mathcal{N}} \Delta_i \, p(i, C). \tag{1}$$

Assumption 2 (Full support)**.** For every $x \in \{0, 1\}^N$, $P[(X_1, \dots, X_N) = x] > 0$.

For comparison we consider a *planner's dictatorship*: an outcome $Z_q \sim \text{Bernoulli}(q)$ independent of $(X_1, \dots, X_N)$, giving welfare $V(q) = \sum_{i \in \mathcal{N}} \Delta_i \, P[X_i = Z_q]$. The objective is linear in q, so the best dictatorship is either $q = 0$ or $q = 1$.
Definition 1 (High-stake franchise)**.** $C \subseteq \mathcal{N}$ is *high-stake* if $C = \mathcal{N}$ or $\Delta_i \geq \Delta_j$ for every $i \in C$ and every $j \in \mathcal{N} \setminus C$.

3 Two negative results

3.1 Dictatorship can beat every franchise

Proposition 1. *There exist stakes $(\Delta_1, \dots, \Delta_5)$ and a full-support distribution under which every odd franchise has strictly lower expected welfare than the dictatorship $q = 1$.*

Proof. Take $N = 5$ and $(\Delta_1, \dots, \Delta_5) = (5, 4, 3, 2, 1)$. Define P_0 on $\{0, 1\}^5$ by the following four atoms, with integer weights summing to 6:

x	$6\,P_0[X=x]$
$(0,1,1,1,0)$	1
$(1,0,0,1,1)$	3
$(1,0,1,0,1)$	1
$(1,1,0,0,0)$	1.

The dictatorship $q=1$ has welfare $17/2$. The welfares of all sixteen odd franchises, multiplied by 6, are:

C	$6\,U_{P_0}(C)$	C	$6\,U_{P_0}(C)$
$\{1\}$	48	$\{1,2,3\}$	48
$\{2\}$	45	$\{1,2,4\}$	48
$\{3\}$	45	$\{1,2,5\}$	48
$\{4\}$	45	$\{1,3,4\}$	48
$\{5\}$	45	$\{1,3,5\}$	45
$\{1,4,5\}$	45	$\{2,3,4\}$	42
$\{2,3,5\}$	45	$\{2,4,5\}$	45
$\{3,4,5\}$	48	$\{1,2,3,4,5\}$	48.

The best franchise welfare is 8, so the dictatorship beats every franchise by at least $1/2$.

Mix P_0 with the uniform distribution R on $\{0,1\}^5$ to enforce full support:

$$P = \tfrac{99}{100}\,P_0 + \tfrac{1}{100}\,R.$$

Total stake is 15, so replacing P_0 by P can shrink any welfare gap by at most $15/100$. Since

$$\tfrac{99}{100}\cdot\tfrac{1}{2} \;-\; \tfrac{15}{100} \;=\; \tfrac{69}{200} \;>\; 0,$$

the dictatorship $q=1$ still beats every franchise under the full-support distribution P. □

3.2 Optimal franchises need not be high-stake

Proposition 2. *There exist strictly decreasing stakes and a full-support distribution under which the unique optimal franchise is not high-stake.*

Proof. Take $N=5$ and $(\Delta_1,\ldots,\Delta_5)=(5,4,3,2,1)$. Define P_0 on three states:

x	$P_0[X=x]$
$(1,0,0,0,1)$	$1/5$
$(1,0,0,1,1)$	$3/5$
$(1,1,0,0,0)$	$1/5.$

Let $C^\star = \{1,2,4\}$. The welfares $5\,U_{P_0}(C)$ across the sixteen odd franchises are:

C	$5\,U_{P_0}(C)$	C	$5\,U_{P_0}(C)$
$\{1\}$	39	$\{1,2,3\}$	39
$\{2\}$	39	$\{1,2,4\}$	42
$\{3\}$	36	$\{1,2,5\}$	39
$\{4\}$	39	$\{1,3,4\}$	39
$\{5\}$	36	$\{1,3,5\}$	36
$\{1,4,5\}$	36	$\{2,3,4\}$	36
$\{2,3,5\}$	36	$\{2,4,5\}$	39
$\{3,4,5\}$	39	$\{1,2,3,4,5\}$	39.

$C^\star$ is the unique maximizer, beating every other franchise by at least 3/5.

Mixing with the uniform R on $\{0,1\}^5$,

$$P = \tfrac{99}{100}\,P_0 + \tfrac{1}{100}\,R,$$

gives full support. Total stake is 15, so for any other franchise D,

$$U_P(C^\star) - U_P(D) \;\ge\; \tfrac{99}{100}\cdot\tfrac{3}{5} \;-\; \tfrac{1}{100}\cdot 15 \;=\; \tfrac{111}{250} \;>\; 0,$$

and $C^\star$ remains the unique optimum. Because the stakes are strictly decreasing, the only high-stake franchises of odd size are $\{1\}$, $\{1,2,3\}$, and $\{1,2,3,4,5\}$. Since $C^\star$ excludes agent 3 (stake 3) while including agent 4 (stake 2), $C^\star$ is not high-stake. □

Intuition. In Proposition 2, agent 3's opinion is constantly 0 under P_0 and so conveys no information, while agent 4's opinion marks the high-probability state in which outcome 1 best serves the high-stake agents. Swapping voter 3 for voter 4 in the size-three franchise changes Y in exactly the state $(1,0,0,1,1)$, flipping it from 0 (pleasing stakes $4+3=7$) to 1 (pleasing stakes $5+2+1=8$); the net gain $1\times 3/5$ is the entire margin. In Proposition 1, the joint distribution is biased towards $X_1 = 1$, and the constant dictator who sides with the highest-stake agent outperforms any majority of a fluctuating subset.

4 Positive results under iid opinions

The counterexamples exploit two features the literature has long flagged as the difficult cases: outcome asymmetry (used in Proposition 1) and heterogeneous marginal opinions (used in Proposition 2). The next assumption removes both.

Assumption 3 (Common iid opinions)**.** There is $p \in (0,1)$ such that $X_1,\dots,X_N$ are independent Bernoulli random variables with common parameter p.

Assumption 4 (Balanced iid opinions)**.** Assumption 3 holds with $p = 1/2$.

Assumption 3 implies full support; Assumption 4 adds neutrality between the two outcomes.

4.1 Insider advantage

Lemma 1 (Insider advantage)**.** *Under Assumption 3, for every odd $m = 2h + 1$ there exist $\alpha_m, \beta_m \in (0, 1)$, depending only on m and p, such that for every franchise C of size m,*

$$P[X_i = Y_C] \;=\; \begin{cases} \alpha_m & \textit{if } i \in C, \\ \beta_m & \textit{if } i \notin C. \end{cases}$$

Moreover,

$$\alpha_m - \beta_m \;=\; 2\,p(1-p)\,P[B_{m-1} = h] \;>\; 0,$$

where $B_{m-1} \sim \mathrm{Binomial}(m-1, p)$.

Proof. The first claim is iid symmetry. For the difference: fix $i \in C$, let $T = \sum_{j \in C \setminus \{i\}} X_j \sim \mathrm{Binomial}(2h, p)$. Agent i agrees with the majority of C iff $X_i = 1$ and $T \geq h$, or $X_i = 0$ and $T \leq h$, so

$$\alpha_m \;=\; p\,P[T \geq h] \;+\; (1-p)\,P[T \leq h].$$

For an outsider $k \notin C$, set $S = \sum_{j \in C} X_j$. Then

$$\beta_m \;=\; p\,P[S \geq h+1] \;+\; (1-p)\,P[S \leq h].$$

Writing $S = T + X_i$,

$$\begin{aligned} P[S \geq h+1] \;&=\; p\,P[T \geq h] + (1-p)\,P[T \geq h+1], \\ P[S \leq h] \;&=\; (1-p)\,P[T \leq h] + p\,P[T \leq h-1]. \end{aligned}$$

Set $c = P[T = h]$. Then $P[T \geq h] - P[S \geq h+1] = (1-p)\,c$ and $P[T \leq h] - P[S \leq h] = p\,c$, so

$$\alpha_m - \beta_m \;=\; p(1-p)\,c \;+\; (1-p)\,p\,c \;=\; 2\,p(1-p)\,c \;>\; 0.$$ □

4.2 High-stake optimality

Theorem 1. *Under Assumption 3, every optimal franchise is high-stake.*

Proof. For fixed odd m, Lemma 1 gives

$$U(C) \;=\; \beta_m \sum_{i \in \mathcal{N}} \Delta_i \;+\; (\alpha_m - \beta_m) \sum_{i \in C} \Delta_i.$$

The first term does not depend on C and $\alpha_m - \beta_m > 0$, so among franchises of size m welfare is maximized by enfranchising the m highest-stake agents, up to ties. Such franchises are high-stake. The feasible odd sizes are finite, so a globally optimal size exists, and at that size the same fixed-size argument forces every optimum to be high-stake. □

4.3 Voting beats dictatorship under balanced opinions

Theorem 2. *Under Assumption 4, every odd franchise C satisfies $U(C) \geq V(q)$ for every $q \in [0,1]$, with strict inequality whenever C contains a positive-stake agent.*

Proof. With $X_i \sim \text{Bernoulli}(1/2)$ and Z_q independent of all opinions, $P[X_i = Z_q] = 1/2$ for every i and q, so every planner's dictatorship has welfare $V(q) = \frac{1}{2}\sum_{i\in\mathcal{N}} \Delta_i$. For an odd franchise C of size m, an outsider's vote is independent of Y_C, so $\beta_m = 1/2$. Lemma 1 gives $\alpha_m > 1/2$, so

$$U(C) \;=\; \tfrac{1}{2}\sum_{i\in\mathcal{N}} \Delta_i \;+\; \big(\alpha_m - \tfrac{1}{2}\big)\sum_{i\in C} \Delta_i \;\geq\; \tfrac{1}{2}\sum_{i\in\mathcal{N}} \Delta_i \;=\; V(q). \qquad \square$$

4.4 A prefix rule for fair-coin opinions

For odd m, write

$$D_m \;=\; \sum_{i=1}^{m} \Delta_i, \qquad a_m \;=\; \alpha_m - \tfrac{1}{2} \;=\; \frac{1}{2^m}\binom{m-1}{(m-1)/2}.$$

Theorem 3. *Under Assumption 4, an optimal franchise is the prefix $C_m = \{1,\dots,m\}$ for any odd m that maximizes $a_m\, D_m$.*

Proof. By Theorem 1, for fixed odd m an optimal franchise of size m is the prefix C_m, up to ties. Substituting $\beta_m = 1/2$ into the welfare formula from the proof of Theorem 1 gives

$$U(C_m) \;=\; \tfrac{1}{2}\sum_{i\in\mathcal{N}} \Delta_i \;+\; a_m\, D_m.$$

The first term is independent of m, so the optimal odd size maximizes $a_m\, D_m$. $\square$

Corollary 1. *Under Assumption 4, if $\Delta_i = \delta > 0$ for every i, the full franchise $\mathcal{N}$ is the unique optimum.*

Proof. With $\Delta_i = \delta$, the objective is $a_m D_m = \delta m\, a_m$. Writing $m = 2r+1$, the ratio at consecutive odd sizes is

$$\frac{(m+2)\, a_{m+2}}{m\, a_m} \;=\; \frac{2r+3}{2r+2} \;>\; 1.$$

So $a_m D_m$ strictly increases in odd m, and the largest feasible odd size is N. $\square$

Where the assumptions bite. Theorem 2 needs balance: the dictatorship counterexample puts most mass on states with $X_1 = 1$, and dictating $q = 1$ exploits exactly that asymmetry. The common-distribution part of Assumption 3 is what powers Theorem 1: when agents have heterogeneous marginals, an enfranchised low-stake agent may be valuable because her vote tracks high-stake outsiders, which is the force in the counterexample of Proposition 2. This is also why the expert-voting literature [2, 3] ranks voters by competence rather than by welfare stake.

References

[1] Franz Dietrich and Kai Spiekermann. Jury Theorems. *Stanford Encyclopedia of Philosophy*, 2021. `https://plato.stanford.edu/entries/jury-theorems/`.

[2] Shmuel Nitzan and Jacob Paroush. Optimal decision rules in uncertain dichotomous choice situations. *International Economic Review*, 23(2):289–297, 1982.

[3] Daniel Berend, Luba Sapir, and Shmuel Sapir. Ranking of weighted majority rules. *Journal of Applied Probability*, 45(4):994–1006, 2008.

[4] Patrick W. Schmitz and Thomas Tröger. The (sub-)optimality of the majority rule. *Games and Economic Behavior*, 74(2):651–665, 2012.

[5] L. S. Penrose. The elementary statistics of majority voting. *Journal of the Royal Statistical Society*, 109:53–57, 1946.

[6] John F. Banzhaf III. Weighted voting doesn't work: a mathematical analysis. *Rutgers Law Review*, 19:317–343, 1965.

[7] Alessandra Casella and Andrew Gelman. A simple scheme to improve the efficiency of referenda. *Journal of Public Economics*, 92(10–11):2240–2261, 2008.